Distinguishing between relations and functions

1 What is the name used to describe a graph where for some value of x, there exists two or more different values of y?

2 Determine whether the following statements are true or false:

- **a** When working with a function, substituting a certain value of x into the formula gives only 1 value of y for that value of x.
- **b** A horizontal line can intersect the graph of a function at more than one point.
- **c** A relation always passes the vertical line test.
- **d** All functions are relations.

3 State whether each of the following statements are correct:

- **a** Some relations are functions.
- **b** All functions are relations.
- **c** No functions are relations.
- **d** No relations are functions,

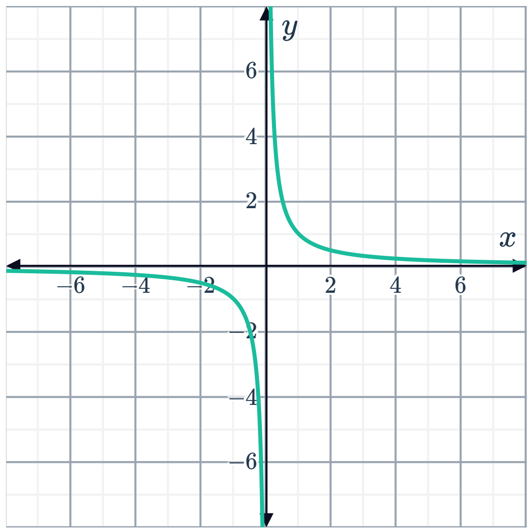
4 State whether each of the following statements are correct:

- **a** The graph of every non-vertical straight line is a function.
- **b** There is no straight line graph that is a function.
- **c** The graph of every straight line is a relation.
- **d** The graph of every non-horizontal straight line is a function.

5 Determine whether each of the following

- a function,
- a relation,
- a relation and a function, or
- neither

a



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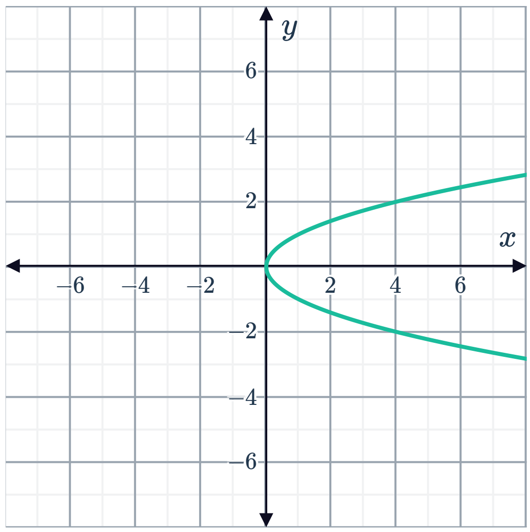

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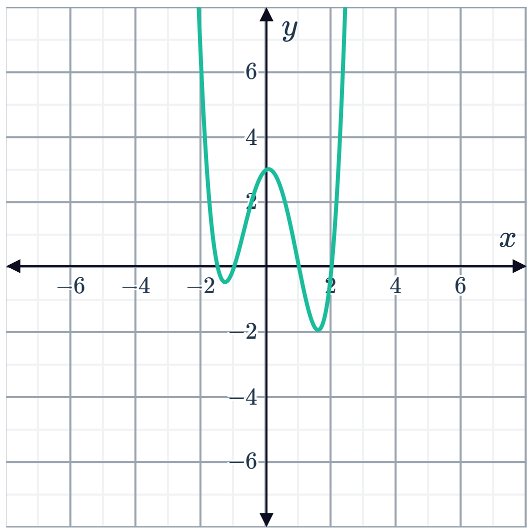

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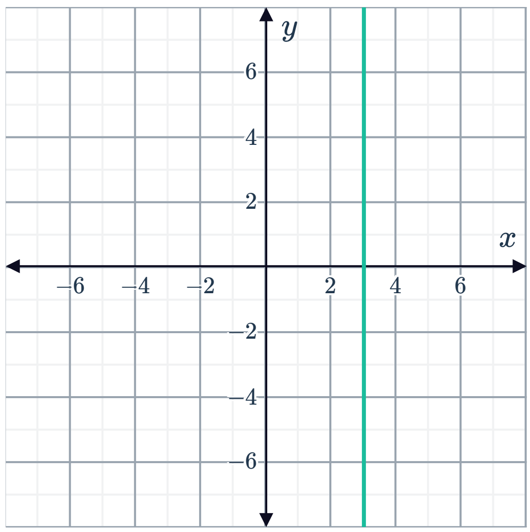

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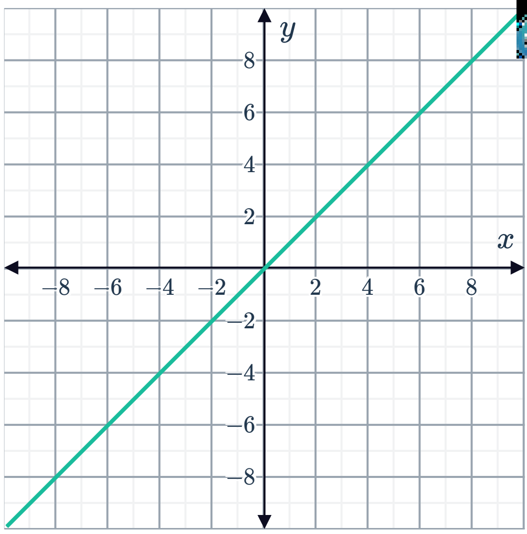

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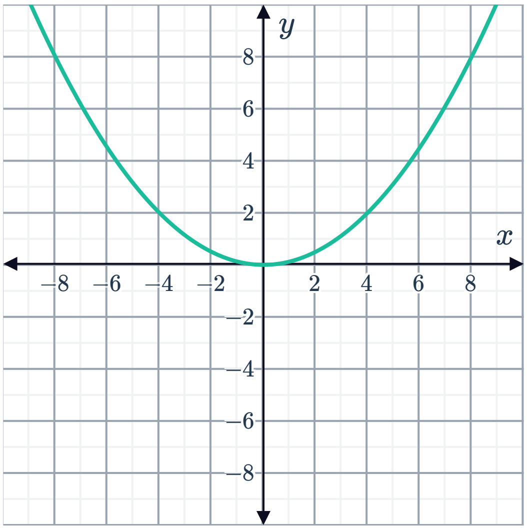

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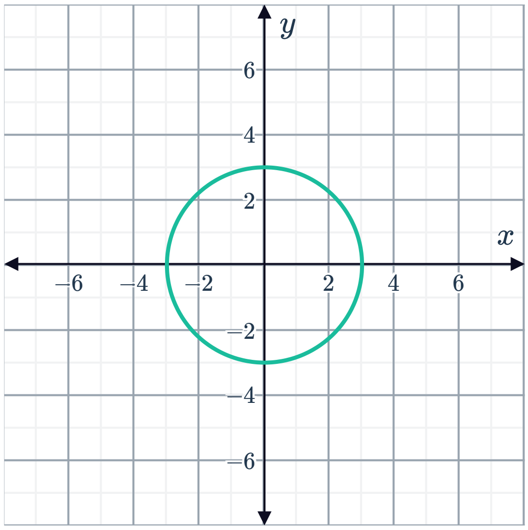

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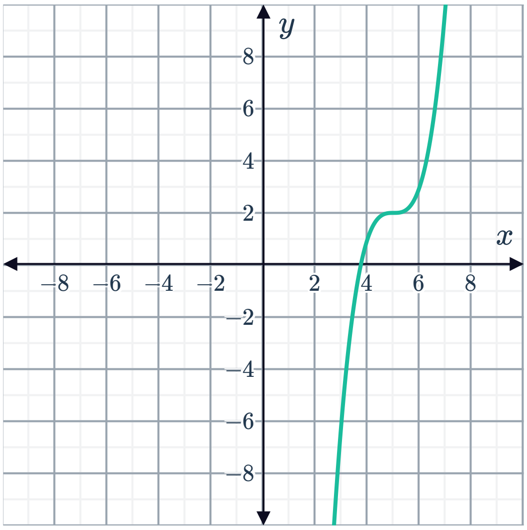

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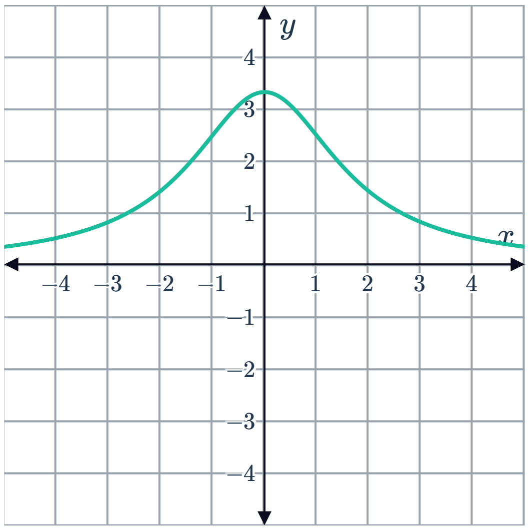

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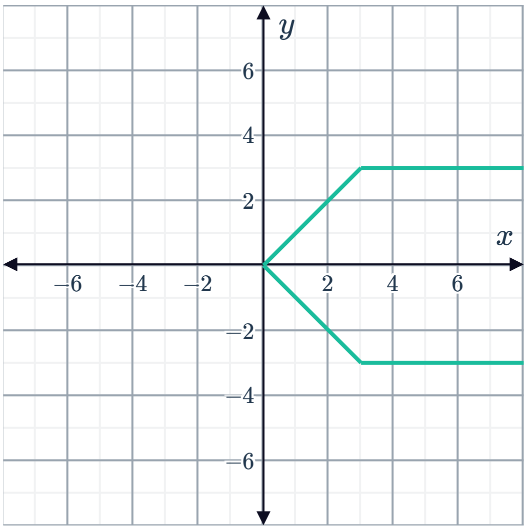

k $\{(2,5),(7,-3),(5,2),(-4,-9)\}$

l $\{(2,5),(2,7),(-3,-4),(-9,13)\}$

m $(1,5),(1,1),(7,-2),(-5,-10)$

n $(1,5),(7,-2),(-5,-10),(13,-13)$

o $(1,5),(1,7),(-2,-5),(-5,-10)$

p $y=9x$

q $y=x^2+2$

r $y=\dfrac{3}{x}$

u $y = -x^3$

6 State whether the values in the tables below represent a function:

a

x	-4	-3	-2	-1	0	1	2	3	4
y	-4	-3	-2	-1	0	-1	-2	-3	-4

b

x	0	1	4	8	9	12	16	18	20
y	0	1	2	$\frac{2}{\sqrt{2}}$	3	$\frac{2}{\sqrt{3}}$	4	$\frac{3}{\sqrt{2}}$	$\frac{2}{\sqrt{5}}$

c

x	-9	-7	-6	-5	-3	-2	3	5	10
y	10	10	10	10	10	10	10	10	10

d

x	-3	-2	-2	0	1	1	2
y	8	4	3	7	2	1	0

7 Consider the points in the table.

x	-4	-3	-2	-1	0	1	2	3	4
y	4	3	2	1	0	1	2	3	4

a Plot the points on a number plane.

b Do they represent a function?

c Write down the coordinates of all the points that would need to be removed so that the remaining points form an increasing function.

8 Consider the points in the table:

x	-4	-3	0	-1	0	1	2	3	4
y	4	3	2	1	0	1	2	3	4

a Plot the points on a number plane.

b Do they represent a function?

9 A relation is defined as follows:

$y=1$ if x is positive and $y=-1$ if x is 0 or negative.

a Complete the table for this relation.

x	-4	-3	-2	-1	0	1	2	3	4
y									

b Plot the points on a number plane.

c Do these values represent a function?

10 Consider these ordered pairs:

$\{(-9,-5),(-5,-10),(-5,-4),(-3,7),(-2,-4),(-1,1)\}$

a Plot the ordered pairs on a number plane.

b Which ordered pair would need to be removed from the set so that the remaining ordered pairs represent a function?

11 Write down a value of k such that the relation $\{(6,2),(8,5),(1,7),(k,4)\}$ does not represent a function.

12 Does there exist any values of k such that the relation $\{(2,1),(7,k),(5,8),(3,9)\}$ does not represent a function?

Applications

13 CheapCalls Mobile charges $\$1.10$ a minute plus a connection fee of $70c$ for an international call.

a Complete the given table:

b Is this relation a function?

Call length (minutes)	International Call cost (dollars)
1	
2	
3	
4	
5	

14 A particular store is offering one free T-shirt for every two T-shirts purchased, where each T-shirt costs $\$19$.

a Complete the table:

b Is this relation a function?

Number of T-shirts	Total Cost (dollars)
1	
2	
3	
4	
5	

15 A particular internet service provider charges $\$40$ a month on their 20 GB plan and charges $\$10$ for each additional 10 GB used.

a Complete the given:

b Is this relation a function?

Total GB Used	Total Charge (dollars)
10	
20	
30	
40	
50	